



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0905 JAVA PROGRAMMING – C01 – ASSESSMENT 3

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1.

The screenshot displays the SIMATS Online Programming Lab interface. The top header shows the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" with "JAVA PROGRAMMING" below it. The user's name "KARANYA S" and ID "192421309RA" are visible in the top right corner.

The main area is divided into three sections:

- Your Code:** A text editor showing Java code for a program that reads employee details and calculates their salary. The code includes imports, a Scanner, and arrays for employees. It handles different employee types like TeachingFaculty, TechnicalStaff, and VisitingFaculty.
- Input:** A text area containing the input data:

```
3
TeachingFaculty
101
Dr.Arun
```
- Output:** A text area showing the program's output:

```
Employee Details
Employee ID : 101
Name : Dr.Arun
Department : CSE
Monthly Salary : 60000.0
Employee ID : 102
```

Below the output, there are three test cases, each with a red 'X' icon, indicating they have failed. The first test case is labeled "Test Case 1", the second "Test Case 2", and the third "Test Case 3".

The bottom section of the screenshot shows the continuation of the code and the test cases. The code continues with the calculation of the total salary and the highest salary. The test cases are still marked as failed, with "0.00 / 25 marks" displayed.

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KARANYA S 192421309RA

```
44 }
45 System.out.println("Highest Paid Employee");
46 System.out.println("Employee ID : "+highest.id);
47 System.out.println("Salary : "+highest.calculateSalary());
48 System.out.println("Average Salary");
49 System.out.printf("%.2f\n",total/n);
50 for(Employee e:emp){
51     if(e.id==searchId){
52         System.out.println("Employee Found");
53         System.out.println("Employee ID : "+e.id);
54         System.out.println("Name : "+e.name);
55         System.out.println("Salary : "+e.calculateSalary());
56         break;
57     }
58 }
59 }
60 }
61 class Employee{
62     int id;
63     String name,dept;
64     double basic;
65     Employee(int id,String name,String dept,double basic){
```

9

Windows Taskbar: File Explorer, Edge, VS Code, HP, etc.

System Tray: ENG IN, 38%, 09:05 PM, 28-07-2026

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```
75 class TeachingFaculty extends Employee{
76     int hours;
77     double incentive;
78     TeachingFaculty(int id,String name,String dept,double basic,int
79         super(id,name,dept,basic);
80         this.hours=hours;
81         this.incentive=incentive;
82     }
83     @Override
84     double calculateSalary(){
85         return basic+(hours*incentive);
86     }
87 }
88 class TechnicalStaff extends Employee{
89     int overtime;
90     double rate;
91     TechnicalStaff(int id,String name,String dept,double basic,int
92         super(id,name,dept,basic);
93         this.overtime=overtime;
94         this.rate=rate;
95     }
96     @Override
```

9

Windows Taskbar: File Explorer, Edge, VS Code, HP, etc.

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```
95 }
96 @Override
97 double calculateSalary(){
98     return basic+(overtime*rate);
99 }
100 }
101 class VisitingFaculty extends Employee{
102     int lectures;
103     double payment;
104     VisitingFaculty(int id,String name,String dept,double basic,int
105         super(id,name,dept,basic);
106     this.lectures=lectures;
107     this.payment=payment;
108 }
109 @Override
110 double calculateSalary(){
111     return lectures*payment;
112 }
113 }
```

Run Save Submit

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2.

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```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=Integer.parseInt(sc.nextLine());
6         Vehicle[] v=new Vehicle[n];
7         for(int i=0;i<n;i++){
8             String type=sc.nextLine();
9             String number=sc.nextLine();
10            String owner=sc.nextLine();
11            double amount=Double.parseDouble(sc.nextLine());
12            if(type.equalsIgnoreCase("Car"))
13                v[i]=new Car(number,owner,amount);
14            else if(type.equalsIgnoreCase("Bike"))
15                v[i]=new Bike(number,owner,amount);
16            else
17                v[i]=new Truck(number,owner,amount);
18        }
19        String search=sc.nextLine();
20        double max=0;
21        Vehicle highest=null;
22        for(Vehicle x:v){
```

3
Car
TN18AB1234
Ramesh

Output

Vehicle:TN18AB1234
Claim Amount:640000.0

Vehicle:TN09XY5678
Claim Amount:60000.0

Visible Test Cases

Test Case 1 ✖

Test Case 2 ✖

Test Case 3 ✖

0.00 / 25 marks

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```
67 super(vehicleNumber,ownerName,insuredAmount);
68 }
69 double calculateClaimAmount() {
70     return insuredAmount*0.60;
71 }
72 }
73 class Truck extends Vehicle{
74     Truck(String vehicleNumber,String ownerName,double insuredAmount)
75     {
76         super(vehicleNumber,ownerName,insuredAmount);
77     }
78     double calculateClaimAmount() {
79         return insuredAmount*0.90;
80     }
81 }
```

Run Save Submit

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```
57 class Car extends Vehicle{
58     Car(String vehicleNumber,String ownerName,double insuredAmount){
59         super(vehicleNumber,ownerName,insuredAmount);
60     }
61     double calculateClaimAmount() {
62         return insuredAmount*0.80;
63     }
64 }
65 class Bike extends Vehicle{
66     Bike(String vehicleNumber,String ownerName,double insuredAmount)
67     {
68         super(vehicleNumber,ownerName,insuredAmount);
69     }
70     double calculateClaimAmount() {
71         return insuredAmount*0.60;
72     }
73 }
74 class Truck extends Vehicle{
75     Truck(String vehicleNumber,String ownerName,double insuredAmount)
76     {
77         super(vehicleNumber,ownerName,insuredAmount);
78     }
79     double calculateClaimAmount() {
80         return insuredAmount*0.90;
81     }
82 }
```

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```
40 }
41     sc.close();
42 }
43 }
44 class Vehicle{
45     String vehicleNumber,ownerName;
46     double insuredAmount;
47     Vehicle(String vehicleNumber,String ownerName,double insuredAmount){
48         this.vehicleNumber=vehicleNumber;
49         this.ownerName=ownerName;
50         this.insuredAmount=insuredAmount;
51     }
52
53     double calculateClaimAmount() {
54         return insuredAmount;
55     }
56 }
57 class Car extends Vehicle{
58     Car(String vehicleNumber,String ownerName,double insuredAmount){
59         super(vehicleNumber,ownerName,insuredAmount);
60     }
61     double calculateClaimAmount() {
```

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```
19 String search=sc.nextLine();
20 double max=0;
21 Vehicle highest=null;
22 for(Vehicle x:v){
23     System.out.println("\nVehicle:"+x.vehicleNumber);
24     System.out.println("Claim Amount:"+x.calculateClaimAmount());
25     if(x.calculateClaimAmount()>max){
26         max=x.calculateClaimAmount();
27         highest=x;
28     }
29 }
30 System.out.println("\nHighest Claim Vehicle");
31 System.out.println(highest.vehicleNumber);
32 System.out.println(highest.calculateClaimAmount());
33 for(Vehicle x:v){
34     if(x.vehicleNumber.equals(search)){
35         System.out.println("\nVehicle Found");
36         System.out.println("Owner:"+x.ownerName);
37         System.out.println("Claim Amount:"+x.calculateClaimAmount());
38         break;
39     }
40 }
```

Test Case 3

0.00 / 25 marks